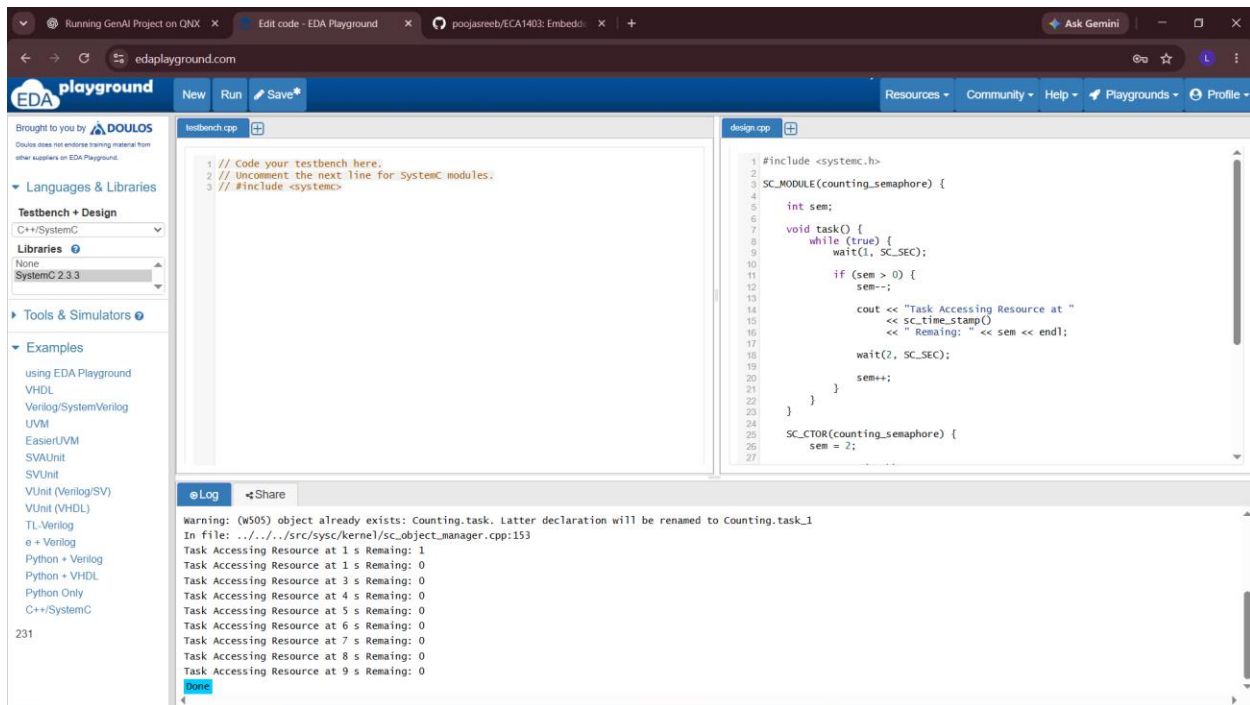




SOURCE CODE

```
#include <systemc.h>
```

```
SC_MODULE(counting_semaphore) {
```

```
    int sem;
```

```
    void task() {
        while (true) {
            wait(1, SC_SEC);
```

```
            if (sem > 0) {
                sem--;
```

```
                cout << "Task Accessing Resource at "
                     << sc_time_stamp()
                     << " Remaining: " << sem << endl;
```

```
                wait(2, SC_SEC);
```

```
                sem++;
```

```
            }
        }
    }
```

```
SC_CTOR(counting_semaphore) {
```

```
    sem = 2;

    SC_THREAD(task);
    SC_THREAD(task);
    SC_THREAD(task);
}
};

int sc_main(int argc, char* argv[]) {

    counting_semaphore obj("Counting");

    sc_start(10, SC_SEC);

    return 0;
}
```